

Inverse Functions (Part II) Homework

Find the equation of the inverse function.

① $f(x) = 5^{x-1} + 2$

② $f(x) = -8(7^{x+1}) - 5$

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③ $f(x) = \log_9(1 - \frac{x}{4})$

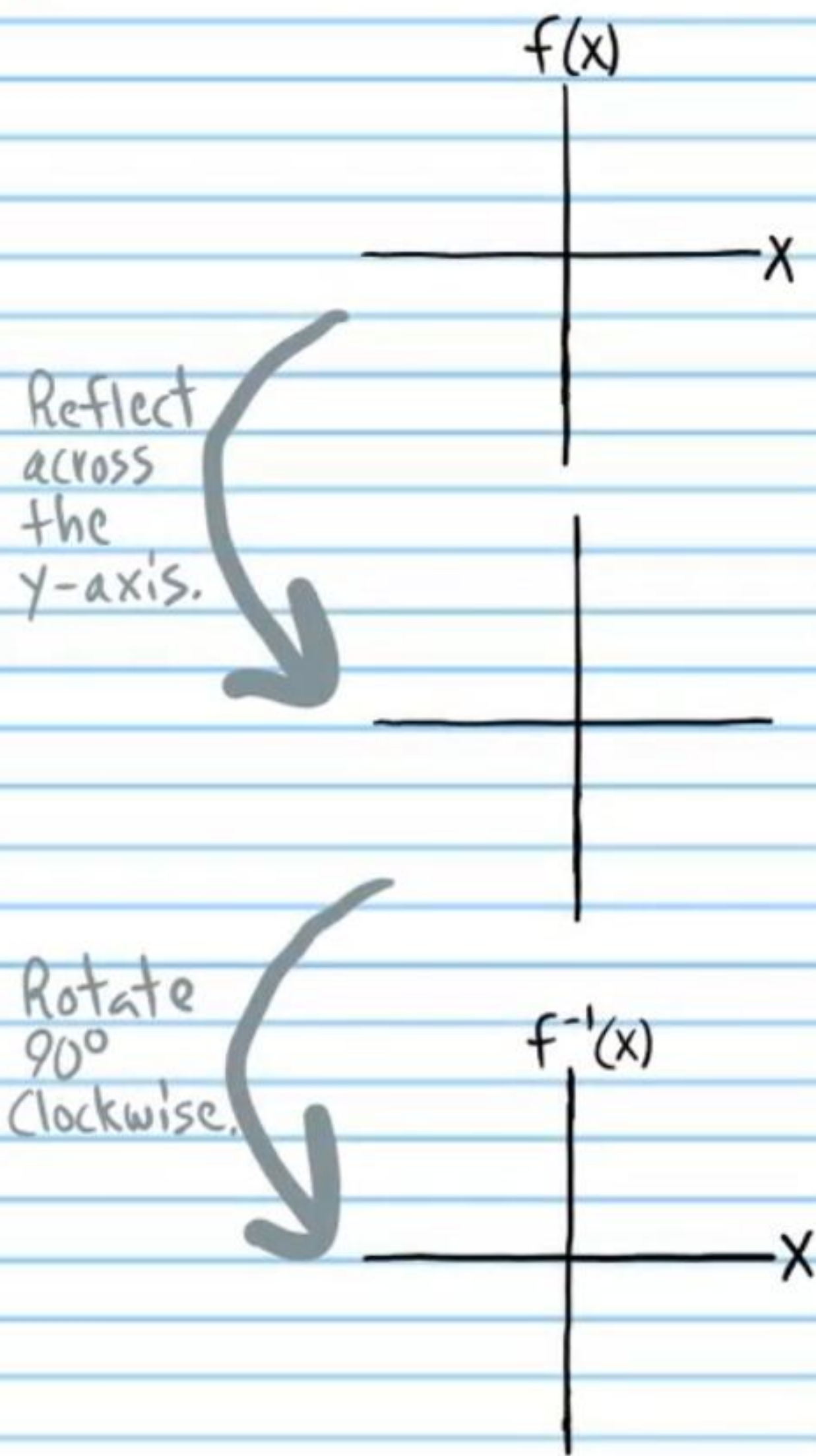
④ $f(x) = \frac{\log_3 x}{-7} + 10$

$$\textcircled{5} f(x) = \frac{3}{x}$$

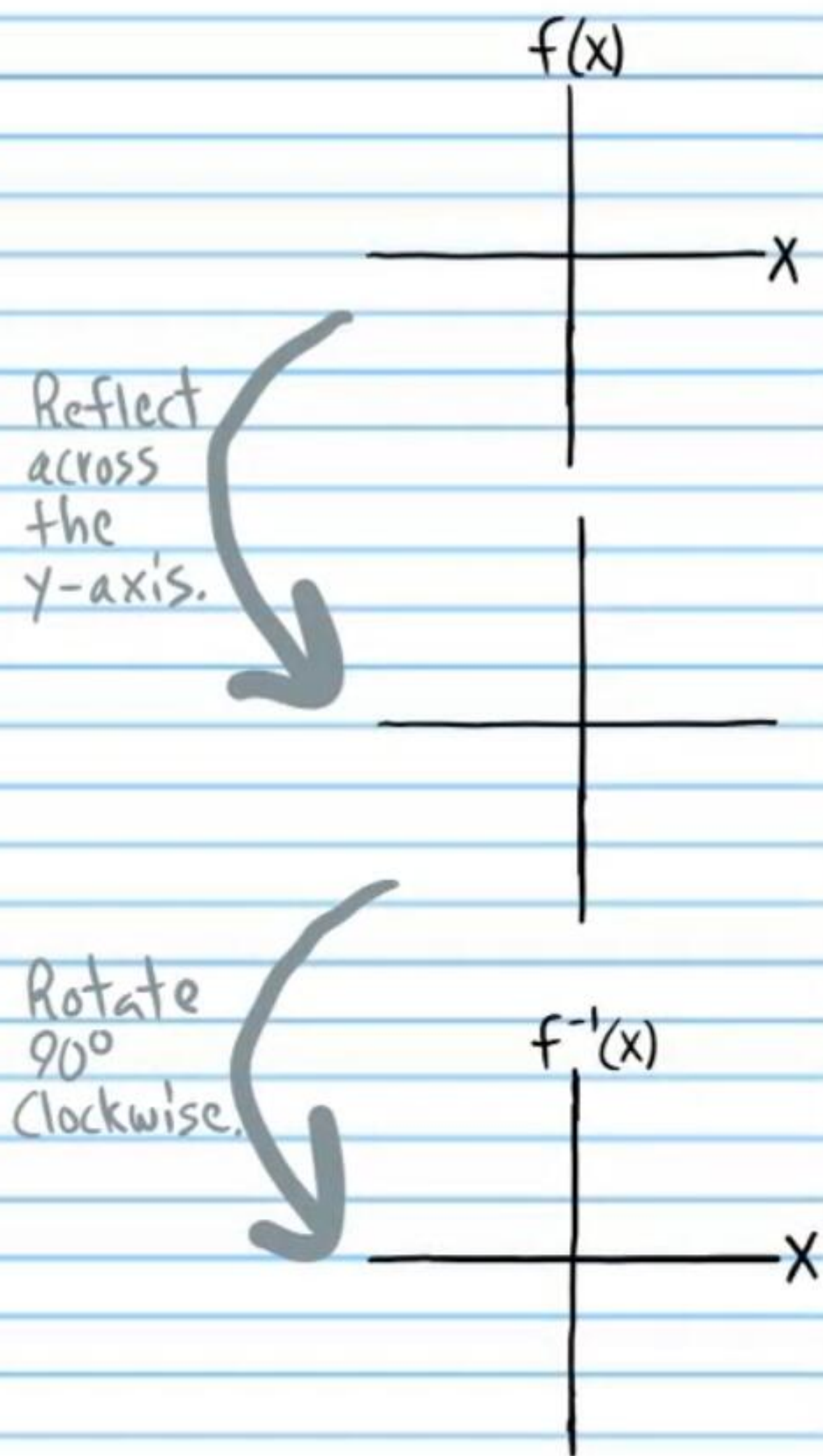
$$\textcircled{6} f(x) = \frac{1}{x-2}$$

In the following problems, restrictions must be used on the original function and/or inverse function so that they are perfectly matched and no aberrations occur. To understand this, graph the function and note that it fails the Horizontal Line Test, which means that it is not one-to-one. Draw the inverse graph and note that it fails the Vertical Line Test. Highlight the parts of the inverse graph and the original graph that you will omit and use inequality notation next to the equations to indicate restrictions.

⑦ $f(x) = \cot x$



⑧ $f(x) = \sec x$



Show algebraically that "f" and "g" are inverse functions.

⑨ $f(x) = \frac{x+2}{4}$ & $g(x) = 4x-2$

⑩ $f(x) = (x-3)^5 - 4$ & $g(x) = \sqrt[5]{x+4} + 3$

Show algebraically that "f" and "f⁻¹" are inverse functions.

⑪ $f(x) = \frac{8^{x+1}}{3}$ & $f^{-1}(x) = \log_8(3x) - 1$

⑫ $f(x) = \frac{\log_2 x - 5}{10}$ & $f^{-1}(x) = 2^{10x+5}$

$$\textcircled{13} \quad f(x) = -\frac{1}{2x} \quad \& \quad f^{-1}(x) = -\frac{1}{2x}$$

$$\textcircled{14} \quad f(x) = \frac{4}{2x-1} \quad \& \quad f^{-1}(x) = \frac{4+x}{2x}$$

Homework Answers

① $f^{-1}(x) = \log_5(x-2) + 1$

② $f^{-1}(x) = \log_7\left(\frac{x+5}{-8}\right) - 1$

③ $f^{-1}(x) = -4(9^x) + 4$

④ $f^{-1}(x) = 3^{-7x+70}$

⑤ $f^{-1}(x) = \frac{3}{x}$

⑥ $f^{-1}(x) = \frac{1}{x} + 2$ or $\frac{1+2x}{x}$

⑦ $f(x) = \cot x, 0^\circ < x < 180^\circ$

$f^{-1}(x) = \cot^{-1} x, 0^\circ < y < 180^\circ$

⑧ $f(x) = \sec x, 0^\circ \leq x \leq 180^\circ, x \neq 90^\circ$

$f^{-1}(x) = \sec^{-1} x, 0^\circ \leq y \leq 180^\circ, y \neq 90^\circ$

⑨ $f(g(x))$

$\frac{g(x)+2}{4}$

$\frac{4x-2+2}{4}$

$\frac{4x}{4}$

x



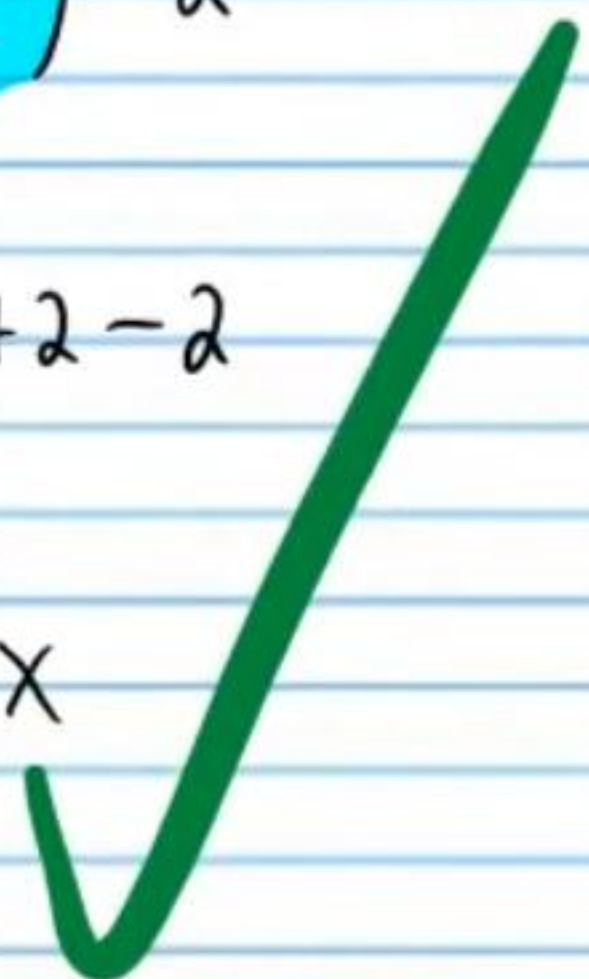
$g(f(x))$

$4f(x)-2$

$4\left(\frac{x+2}{4}\right)-2$

$x+2-2$

x



⑩ $f(g(x))$

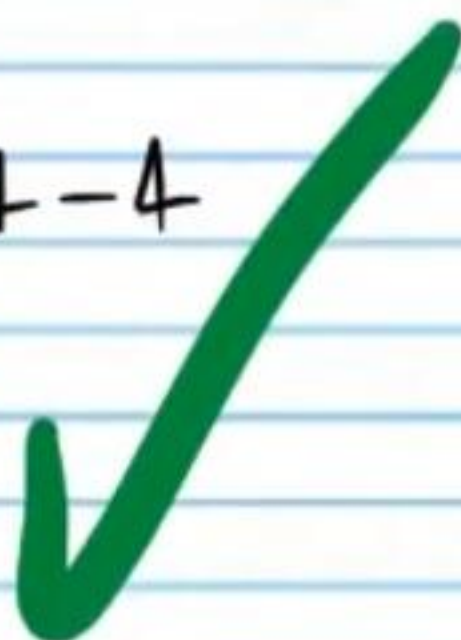
$$(g(x)-3)^5-4$$

$$(\sqrt[5]{x+4}+3-3)^5-4$$

$$(\sqrt[5]{x+4})^5-4$$

$$x+4-4$$

x



$g(f(x))$

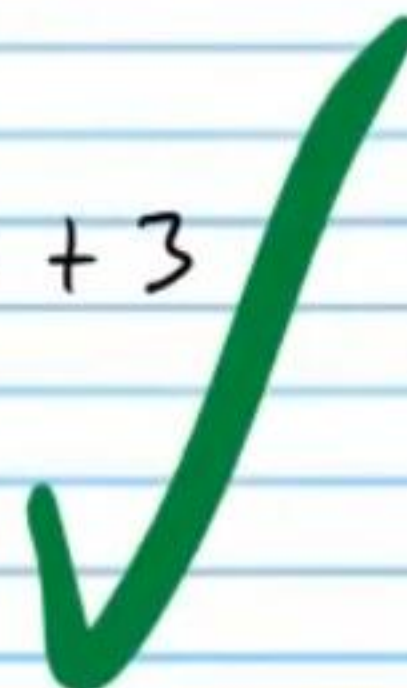
$$\sqrt[5]{f(x)+4}+3$$

$$\sqrt[5]{(x-3)^5-4+4}+3$$

$$\sqrt[5]{(x-3)^5}+3$$

$$x-3+3$$

x



⑪ $f(f^{-1}(x))$

$$\frac{f^{-1}(x)+1}{8}$$

$$\frac{8^{\log_8(3x)-1}+1}{3}$$

$$\frac{8^{\log_8(3x)}}{3}$$

$$\frac{3x}{3}$$

x



$f^{-1}(f(x))$

$$\log_8[3f(x)]-1$$

$$\log_8\left[3\left(\frac{8^{x+1}}{3}\right)\right]-1$$

$$\log_8 8^{x+1}-1$$

$$x+1-1$$

x



⑫ $f(f^{-1}(x))$

$$\frac{\log_2 f^{-1}(x) - 5}{10}$$

$$\frac{\log_2 2^{10x+5} - 5}{10}$$

$$\frac{10x + 5 - 5}{10}$$

$$\frac{10x}{10}$$

x



$f^{-1}(f(x))$

$$2^{10[f(x)]+5}$$

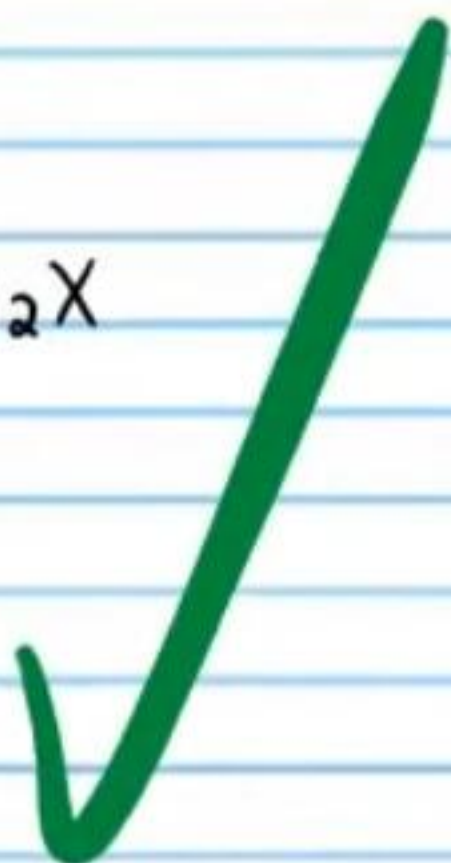
$$2^{10f(x)+5}$$

$$2^{10\left(\frac{\log_2 x - 5}{10}\right)+5}$$

$$2^{\log_2 x - 5 + 5}$$

$$2^{\log_2 x}$$

x



⑬ $f(f^{-1}(x))$

$$-\frac{1}{2f^{-1}(x)}$$

$$-\frac{1}{2\left(-\frac{1}{2x}\right)}$$

$$-\frac{1}{-\frac{1}{x}}$$

$$\frac{1}{\frac{1}{x}}$$

x



$f^{-1}(f(x))$

$$-\frac{1}{2f(x)}$$

$$-\frac{1}{2\left(-\frac{1}{2x}\right)}$$

$$-\frac{1}{-\frac{1}{x}}$$

$$\frac{1}{\frac{1}{x}}$$

x



⑭ $f(f^{-1}(x))$

$$\frac{4}{2f^{-1}(x)-1}$$

$$\frac{4}{2\left(\frac{4+x}{2x}\right)-1}$$

$$\frac{4}{\frac{4+x}{x}-1}$$

$$\frac{4}{\frac{4+x}{x}-\frac{x}{x}}$$

$$\frac{4}{\frac{4+x-x}{x}}$$

$$\frac{4}{\frac{4}{x}}$$

$$x$$

$f^{-1}(f(x))$

$$\frac{4+f(x)}{2f(x)}$$

$$\frac{4+\frac{4}{2x-1}}{2\left(\frac{4}{2x-1}\right)}$$

$$\frac{\frac{4}{1}\frac{(2x-1)}{(2x-1)}+\frac{4}{2x-1}}{\frac{8}{2x-1}}$$

$$\frac{\frac{8x-4}{2x-1}+\frac{4}{2x-1}}{\frac{8}{2x-1}}$$

$$\frac{\frac{8x-4+4}{2x-1}}{\frac{8}{2x-1}}$$

$$\frac{\frac{8x}{2x-1}}{\frac{8}{2x-1}}$$

$$\frac{8x}{2x-1} \div \frac{8}{2x-1}$$

$$\frac{8x}{2x-1} \cdot \frac{2x-1}{8}$$

$$x$$